

A Structure-Agnostic SHM Decision Framework for Reusable CFRP Spacecraft Structures

from Real-Data Detection to Physics-Grounded Life Prognosis

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Keio University / Composites Part B (draft)

The reusable-vehicle decision

Reusable launch vehicles change the SHM question:

- not “did it survive one flight?”
- but **may it fly again — and for how many flights?**

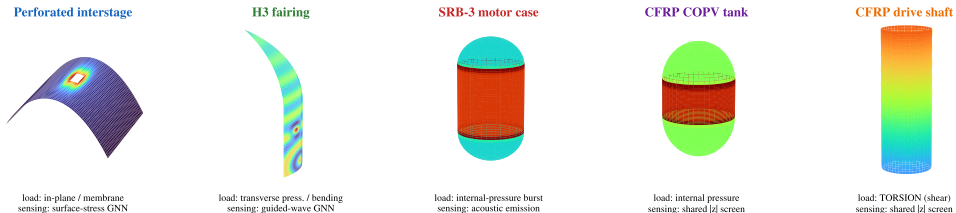
CFRP primary structures (interstage, fairing, motor case, tank, drive shaft) fail by *internal* delamination / debond — invisible, inferred from NDT.

Gap: no end-to-end pipeline that propagates uncertainty from detection to a re-flight decision *and* states what real data supports.

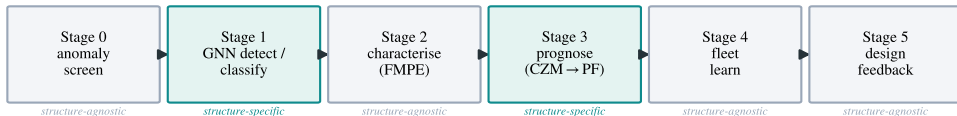
detect → classify → characterise
→ prognose → clear
→ fleet-learn → design

One decision core, five structures, four load modes

A structure-agnostic SHM decision framework across five reusable CFRP spacecraft structures and four load modes



Shared structure-agnostic decision core --- only detection (Stage 1) & prognosis physics (Stage 3) are structure-specific (pluggable)



detect → classify → characterise → prognosis → clear (OK/REPAIR/RETIRE) → fleet-learn → design | 1,499 real-FEM states, one margin surrogate $R^2=0.94$

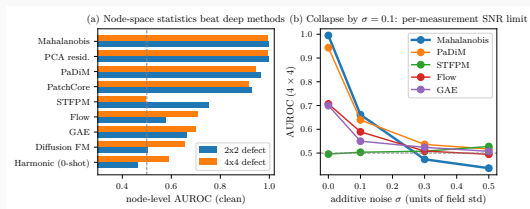
- Shared, structure-agnostic decision core (Stage 0/2/4/5 + clearance).
- Only *detection* (Stage 1) and *prognosis physics* (Stage 3) are structure-specific — pluggable.

The five structures span the load-mode space

structure	load mode	detection front-end	evidence
interstage	membrane	surface-stress GNN	FEM + meas. TSA
fairing	transv./bending	guided-wave GNN	FEM + real OGW
SRB-3 case	pressure burst	acoustic emission	FEM + real AE
drive shaft	torsion	shared $ z $ (future)	CZM-FEM
COPV tank	int. pressure	shared $ z $ (future)	FEM

- Three demonstrated end-to-end with **real data**; two extend the load-mode coverage with FEM prognosis.

Detection front-ends — grounded in real data



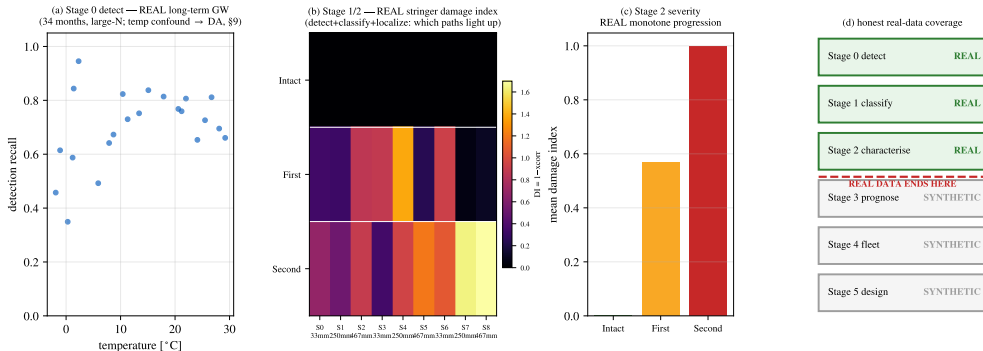
Stage-0 $|z|$ screen: per-node statistic dominates (AUROC ≈ 0.99 , fixed geometry).

Structure-specific Stage-1 detectors:

- interstage: surface-stress GNN, macro-F1 0.76 (vs GAT 0.61); detection generalises, localisation collapses OOD.
- fairing: guided-wave GNN, F1 0.86.
- SRB-3: real AE, per-hit AUROC 0.89 / per-specimen 1.00.

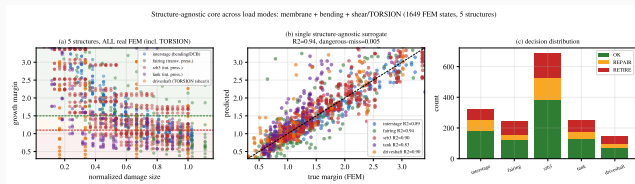
How far does real data carry the pipeline? (honest)

Fairing (CFRP stringer) end-to-end on REAL Open-Guided-Waves data: detect → classify → characterise are real; prognosis is honestly synthetic



- Detect / classify / characterise on the fairing are **real data** (long-term guided waves, impact wavefield).
- **synthetic** from Stage-3 prognosis onward (representative phase-field, synthetic fleet) — the life side is the largest open gap.

Structure-agnostic core — and its boundary

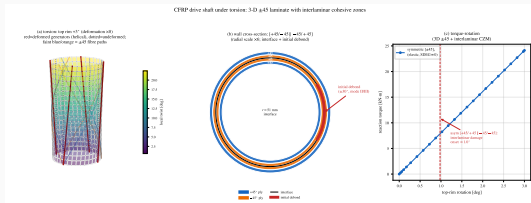


A single margin surrogate over **1,499 real-FEM** critical-load states fits all five ($R^2=0.94$, dangerous-miss 1.1%).

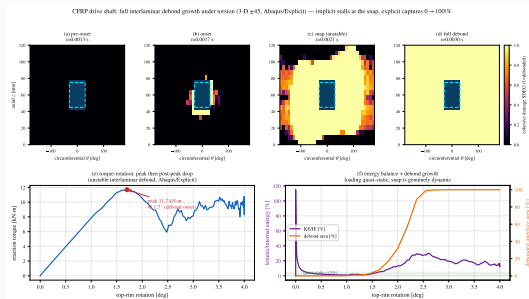
Honest boundary: leave-one-load-mode-out to the unseen torsion shaft *fails* ($R^2 < 0$).

Geometry transfers; load mode does not.

New load mode: the CFRP torsion drive shaft

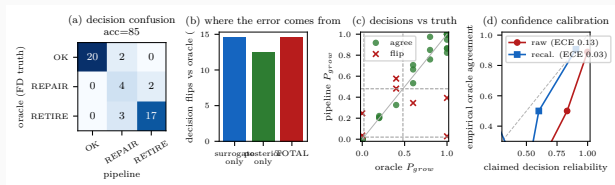


CGAX screen: ± 45 tube, groove damage, T_{cr} sweep.



3-D $\pm 45^\circ$ + interlaminar CZM: full 0 $\rightarrow$ 100% debond.
Implicit stalls at the snap; **explicit** captures it.

End-to-end decision reliability (vs a **synthetic** oracle)

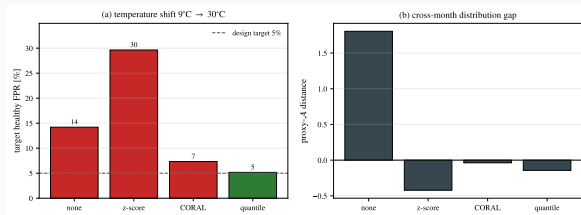


Propagate uncertainty to the clearance call; score against the exact FD-physics oracle.

- accuracy **85%**, dangerous-miss **0%**
- Platt recalibration: ECE 0.18 $\rightarrow$ 0.06

Measured against a **synthetic oracle** — internal error propagation, not agreement with real life tests.

Sim-to-real: normalisation choice controls false alarms

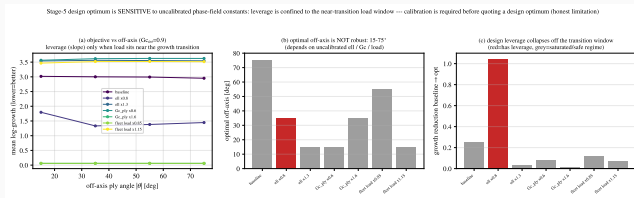


Real OGW temperature shift $9^{\circ} \rightarrow 30^{\circ}\text{C}$:

- uncorrected FPR **14%** (temperature mimics damage)
- quantile DA $\rightarrow$ **5%** (design target)
- **z-score normalisation worsens to 30%**

Interstage TSA: quantile DA closes 97% of the gap.

Stage-5 design feedback — mechanism, not a calibrated value



Fleet-driven inverse design steers off-axis angle + interface toughness.

Honest limitation: under uncalibrated phase-field constants the optimal angle spans 15° – 75° — leverage exists only near the growth transition.

The **mechanism** is robust; a quoted optimum needs calibration.

- **real data**: detection / classification / characterisation (guided waves, AE, TSA), posterior coverage, 1,499-state FEM core.
- **synthetic**: prognosis, clearance, fleet, design — representative phase-field + synthetic fleets; decision UQ vs a synthetic oracle.
- **Largest gap**: no real life-test validation of the life side.
- Priority inputs: Kojima TSA full-field (interstage), real fatigue data.

- The differentiable, fast forward (FNO; autograd-exact) is what a TMCMC / Bayesian-inverse phase needs.
 - Its physics — anisotropic phase-field seeded by interlaminar cohesive zones — is the prognosis layer the torsion shaft extends from mode-I to mode-II/III shear.
- ⇒ **Phase-field** × **GNN prognosis**, calibrated on measured data, is the computational-solid-mechanics continuation.

One decision core across four load modes.

Real where data permits, honest where it does not.

Closing the life-side real-data gap is next.